

Mary Letey

[email](#) + [website](#) + [scholar](#) + [github](#)

Education

Sept 2023 — June 2028	Harvard University	PhD Candidate — Applied Mathematics
Sept 2022 — June 2023	Perimeter Institute	MSc — Theoretical Physics
Sept 2018 — June 2022	University of Cambridge	BA Honours — Mathematics

Publications & Preprints

Sequential correlations change in-context learning: Effective context length and architectural mismatch

[ICML 2026](#) — HiLd Workshop

ML, Lu, Pehlevan, Zavatone-Veth

Pretrain–test task alignment governs generalization in in-context learning

[ICLR 2026](#)

ML, Zavatone-Veth, Lu, Pehlevan

Theory of scaling laws for in-context regression: Depth, width, context and time

[ICLR 2026](#)

Bordelon, **ML**, Pehlevan

Asymptotic theory of in-context learning by linear attention

[Proceedings of the National Academy of Sciences \(2025\)](#)

Lu, **ML**, Zavatone-Veth, Maiti, Pehlevan

Quantum initial conditions for curved inflating universes

[Physical Review D \(2024\)](#)

ML*, Shumaylov*, Agocs, Handley, Hobson, Lasenby

Research & Work Experience

Hudson River Trading — Winter 2026

Research Internship — Machine learning for market prediction

Quebec Artificial Intelligence Institute (MILA) — Summer 2023

Research Internship — Equivariant architectures for neural PDE solvers

Perimeter Institute for Theoretical Physics — 2022-2023

[Master's thesis](#) — Classification of discrete symmetries in non-Euclidean space

Cavendish Laboratory & Kavli Institute for Cosmology, Cambridge — Summer 2022

Research Internship — Research on early-universe cosmology and curved spacetimes

Embryo Ventures — Summer 2020

Internship — Due diligence and founder outreach

Iguana Investments — Fall 2019-Spring 2020

Research Internship — Market prediction models

Invited Talks

EPFL — [FLARE Workshop](#), May 2026

Invited to present on theoretical models of in-context learning

SIAM — Conference on Uncertainty Quantification, March 2026

Invited to present “Asymptotic theory of in-context learning by linear attention”

Kempner Institute — [Spring Into Science Retreat](#), May 2024

Opening talk on “Asymptotic theory of in-context learning by linear attention”

Awards & Funding

2025	Graduate Fellowship	Kempner Institute — Full support for graduate studies
2023	Graduate Prize Fellowship	Harvard University
2022	International Scholar Award	Perimeter Institute — Full support for MSc